Practice Assignment 7 - Not Graded

This assignment will not be graded and is only for practice.
Note:Note: Suppose W_1 and W_2 W1 and W2 are subspaces of a vector space V V . Then sum of two subspaces is defined as:

$$W_1 + W_2 = \{w_1 + w_2 \mid w_1 \in W_1, w_2 \in W_2\}.\text{W1 + W2} = \{\text{w1 + w2} \mid \text{w1} \in \text{W1}, \text{w2} \in \text{W2}\}.$$

One can verify that $W_1 + W_2$ W1 + W2 is a subspace of V V .

Multiple Select Questions (MSQ):

1 point
Let $V_1 = R^3$ V1 = R3 be a vector space, define a function $\|\cdot\|_i : V_1 \to R$ ‖ · ‖i : V1 → R for $i = 0, 1, 2, 3$i = 0, 1, 2, 3 as:

1. $\|(x, y, z)\|_0 = x + y + z$ ‖(x, y, z)‖0 = x + y + z
2. $\|(x, y, z)\|_1 = |x| + |y| + |z|$ ‖(x, y, z)‖1 = |x| + |y| + |z|
3. $\|(x, y, z)\|_2 = \sqrt{x^2 + y^2 + z^2}$ ‖(x, y, z)‖2 = $\sqrt{\text{x2 + y2 + z2}}$
4. $\|(x, y, z)\|_3 = \max\{|x|, |y|, |z|\}$ ‖(x, y, z)‖3 = max{|x|, |y|, |z|}

and let $V_2 = M_{2\times 2}(R)$V2 = M2×2(R) be a vector space, define a function $\|\cdot\|_i : V_2 \to R$ ‖ · ‖i : V2 → R for $i = 4, 5$i = 4, 5 as:

1. $\|A\|_4 = \max\{|a_{11}| + |a_{12}|, |a_{21}| + |a_{22}|\}$ ‖A‖4 = max{|a11| + |a12|, |a21| + |a22|}, where $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$.A = [a11 a12a21 a22] .
2. $\|A\|_5 = \max\{|a_{11}| + |a_{21}|, |a_{12}| + |a_{22}|\}$ ‖A‖5 = max{|a11| + |a21|, |a12| + |a22|}, where $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$A = [a11 a12a21 a22].

Which of the following option(s) is (are) true.

- ☐ $\|\|_0$ ‖‖0 is a norm on V_1 V1.
- ☐ $\|\|_1$ ‖‖1 is not a norm on V_1 V1.
- ☐ $\|\|_2$ ‖‖2 is a norm on V_1 V1.
- ☐ $\|\|_3$ ‖‖3 is not a norm on V_1 V1.
- ☐ $\|\|_4$ ‖‖4 is not a norm on V_2 V2.
- ☐ $\|\|_5$ ‖‖5 is a norm on V_2 V2

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\|\|_2$ $\|\|2$ is a norm on V_1 V1.
$\|\|_5$ $\|\|5$ is a norm on V_2 V2
1 point
Let $\langle .,. \rangle$⟨.,.⟩ denote the standard inner product on R^3 R3, i.e.,
$\langle (x_1, x_2, x_3), (y_1, y_2, y_3) \rangle = x_1 y_1 + x_2 y_2 + x_3 y_3$ ⟨(x1, x2, x3), (y1, y2, y3)⟩ = x1y1 + x2y2 + x3 y3. Consider a linear transformation $T: R^3 \to R$T : R3 → R defined as:

$$T(u) = \langle u, v \rangle, \text{T (u)} = \langle \text{u, v} \rangle,$$

where $v \in R^3$ v ∈ R3. Which of the following options is (are) true for T T?

- [ ] T T is one-one for all $v \neq 0 \in R^3$ v = 0 ∈ R3.
- [ ] T T is onto for all $v \neq 0 \in R^3$ v = 0 ∈ R3.
- [ ] T T is not one-one for every $v \in R^3$ v ∈ R3.
- [ ] There exists a $v \in R^3$ v ∈ R3 such that T T is an isomorphisim.

No, the answer is incorrect.
Score: 0
Accepted Answers:
T T is onto for all $v \neq 0 \in R^3$ v = 0 ∈ R3.
T T is not one-one for every $v \in R^3$ v ∈ R3.
1 point
Which of the following options are true?

- [ ] $V = R^2$ V = R2 and a function defined as:
$$\langle .,. \rangle : V \times V \to R$$
$$\langle (x_1, x_2), (y_1, y_2) \rangle = \sum_{i,j=1}^{2} x_i y_j .$$
⟨.,.⟩ : V × V ⟨(x1, x2), (y1, y2)⟩ → R = i,j=1∑2 xiyj .
is not an inner product on the vector space V V .
- [ ] $V = M_{2\times 2}(R)$V = M2×2(R) and $\langle .,. \rangle_1$ ⟨.,.⟩1 and $\langle .,. \rangle_2$ ⟨.,.⟩2 are two inner products on the vector space V V , a function defined as: $\langle .,. \rangle = \langle .,. \rangle_1 + \langle .,. \rangle_2$ ⟨.,.⟩ = ⟨.,.⟩1 + ⟨.,.⟩2 is an inner product on the vector space V V .
- [ ] $V = M_{3\times 3}(R)$V = M3×3(R) and $\langle .,. \rangle_1$ ⟨.,.⟩1 and $\langle .,. \rangle_2$ ⟨.,.⟩2 are two inner products on the vector space V V , a function defined as: $\langle .,. \rangle = \langle .,. \rangle_1 - \langle .,. \rangle_2$ ⟨.,.⟩ = ⟨.,.⟩1 − ⟨.,.⟩2 is an inner product on the vector space V V .
- [ ] $V = R^2$ V = R2 and a function defined as:
$$\langle .,. \rangle : V \times V \to R$$
$$\langle (x_1, x_2), (y_1, y_2) \rangle = x_1 y_2 - x_2 y_1 .$$
⟨.,.⟩ : V × V ⟨(x1, x2), (y1, y2)⟩ → R = x1y2 − x2 y1.
is an inner product on the vector space V V .

No, the answer is incorrect.
Score: 0
Accepted Answers:
$V = R^2$ V = R2 and a function defined as:

$$\langle .,. \rangle : V \times V \to R$$
$$\langle (x_1, x_2), (y_1, y_2) \rangle = \sum_{i,j=1}^{2} x_i y_j$$. ⟨.,.⟩ : V × V ⟨(x1, x2), (y1, y2)⟩ → R = i,j=1∑2 xiyj .

is not an inner product on the vector space VV .
$V = M_{2\times 2}(R)$V = M2×2(R) and $\langle .,. \rangle_1$ ⟨.,.⟩1 and $\langle .,. \rangle_2$ ⟨.,.⟩2 are two inner products on the vector space VV , a function defined as: $\langle .,. \rangle = \langle .,. \rangle_1 + \langle .,. \rangle_2$ ⟨.,.⟩ = ⟨.,.⟩1 + ⟨.,.⟩2 is an inner product on the vector space VV .
1 point
Consider a linear transformation $T: R^3 \to R^3$ T : R3 → R3 defined as $T(x, y, z) = (x + 2y, x - y + cz, 2x + y + dz)$T(x, y, z) = (x + 2y, x − y + cz, 2x + y + dz). Let AA be the matrix representation of TT with respect to the standard ordered bases of R^3 R3 and $B = [\begin{matrix} 0 & 3 & 2 \\ -1 & -3 & -3 \\ 2 & 3 & 4 \end{matrix}]$B = $\begin{bmatrix} & & \\ & & \\ 0 & -12 & 3-32-34 \end{bmatrix}$.

Based on the above information which of the followings option(s) is(are) true?
☐
If AA is similar to BB, then rank of the linear transformation TT is 2.
☐
If AA is similar to BB, then nullity of the linear transformation TT is 2.
☐
If $c \neq d$c = d, then AA cannot be similar to BB.
☐
If AA is similar to BB and c = 1c = 1, then a basis of the kernel is $\{(\begin{matrix} -2 \\ 1 \\ 3 \end{matrix})\}$. ⌊ . { . ⌈ $\begin{pmatrix} & & \\ & & \\ -2 & 1 & 3 \end{pmatrix}$ ⌋ . } . ⌉

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No, the answer is incorrect.
Score: 0
Accepted Answers:
If AA is similar to BB, then rank of the linear transformation TT is 2.
If $c \neq d$c = d, then AA cannot be similar to BB.

If AA is similar to BB and c = 1c = 1, then a basis of the kernel is $\{(\begin{matrix} -2 \\ 1 \\ 3 \end{matrix})\}$. ⌊ . { . ⌈ $\begin{pmatrix} & & \\ & & \\ -2 & 1 & 3 \end{pmatrix}$ ⌋ . } . ⌉

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1 point
Consider a vector $v = (1, 2)$v = (1, 2) from the inner product space R^2 R2, where inner product is the dot product. Which of the following options is/are true?
☐
There exists a vector u ($\neq v$)u (= v) in R^2 R2 such that $||u|| = ||v||$||u|| = ||v||.
☐
There are infinitely many vectors in R^2 R2 such that norms of the vectors are equal to norm of vv.

☐ There exists a vector uu in R^2 R2 such that $\langle u, v \rangle = 0$.⟨u, v⟩ = 0.

☐ There are infinitely many vectors in R^2 R2 such that angle between the vector and vv is equal to 90° 90∘.

No, the answer is incorrect.

Score: 0

Accepted Answers:

There exists a vector u ($\neq v$)u (≠ v) in R^2 R2 such that $||u|| = ||v||$||u|| = ||v||.

There are infinitely many vectors in R^2 R2 such that norms of the vectors are equal to norm of vv.

There exists a vector uu in R^2 R2 such that $\langle u, v \rangle = 0$.⟨u, v⟩ = 0.

There are infinitely many vectors in R^2 R2 such that angle between the vector and vv is equal to 90° 90∘.

1 point

Let LL and $L^{'}$ L′ be affine subspaces of R^3 R3, where $L = U$L = U and $L^{'} = (2, 0, 1) + U^{'}$ L′ = (2, 0, 1) + U′, for some vector subspaces UU and $U^{'}$ U′ of R^3 R3. Let a basis for UU be given by $\{(2, 0, 1), (1, 1, 0), (0, 1, 0)\}${(2, 0, 1), (1, 1, 0), (0, 1, 0)} and a basis for $U^{'}$ U′ be given by $\{(1, 0, 1), (0, 1, 1)\}${(1, 0, 1), (0, 1, 1)}. Suppose there is a linear transformation $T: U \rightarrow U^{'}$ T : U → U′ such that $(0, 1, 0) \in ker(T)$(0, 1, 0) ∈ ker(T), $T(2, 0, 1) = (0, 1, 1)$T(2, 0, 1) = (0, 1, 1) and $T(1, 1, 0) = (1, 0, 1)$T(1, 1, 0) = (1, 0, 1). An affine mapping $f: L \rightarrow L^{'}$ f : L → L′ is obtained by defining $f(u) = (2, 0, 1) + T(u)$f(u) = (2, 0, 1) + T(u), for all $u \in U$u ∈ U. Which of the following options are true?

☐ $L = \{(x, y, z) \mid x - y - 2z = 0\}$L = {(x, y, z) ∣ x − y − 2z = 0}

☐ $L = R^3$ L = R3

☐ $L^{'} = \{(x, y, z) \mid x + y - z = 1\}$L′ = {(x, y, z) ∣ x + y − z = 1}

☐ $L^{'} = R^3$ L′ = R3

☐ $f(x, y, z) = (x - 2z + 2, z, x - z + 1)$f(x, y, z) = (x − 2z + 2, z, x − z + 1)

☐ $f(x, y, z) = (x - 2z + 2, \frac{x}{2}, x - z + 1)$f(x, y, z) = (x − 2z + 2, 2x, x − z + 1)

☐ $f(x, y, z) = (x - 2z, z, x - z)$f(x, y, z) = (x − 2z, z, x − z)

No, the answer is incorrect.

Score: 0

Accepted Answers:

$L = R^3$ L = R3

$L^{'} = \{(x, y, z) \mid x + y - z = 1\}$L′ = {(x, y, z) ∣ x + y − z = 1}

$f(x, y, z) = (x - 2z + 2, z, x - z + 1)$f(x, y, z) = (x − 2z + 2, z, x − z + 1)

1 point

Let $L_1 = \{(x, y, z) \mid x + y + 2z = 6\}$L1 = {(x, y, z) ∣ x + y + 2z = 6} and $L_2 = \{(x, y, z) \mid 2x + y + 2z = 9\}$L2 = {(x, y, z) ∣ 2x + y + 2z = 9} be affine subspaces of R^3 R3 and W_1 W1 and W_2 W2 be the corresponding vector subspaces of the given affine subspaces of L_1 L1 and L_2 L2 respectively. Choose the set of correct options

☐ $W_1 = \{(x, y, z) \mid x + y + z = 0\}$W1 = {(x, y, z) ∣ x + y + z = 0}

- [ ] $W_2 = \{(x, y, z) \mid 2x + y + 2z = 0\}$W2 = {(x, y, z) | 2x + y + 2z = 0}
- [ ] $W_1 = \{(x, y, z) \mid x + y + 2z = 0\}$W1 = {(x, y, z) | x + y + 2z = 0}
- [ ] $W_2 = \{(x, y, z) \mid 2x + y + 2z = 1\}$W2 = {(x, y, z) | 2x + y + 2z = 1}
- [ ] $W_1 = \{(x, y, z) \mid x + y + 2z = 1\}$W1 = {(x, y, z) | x + y + 2z = 1}
- [ ] $W_1 \cap W_2 = \{(x, y, z) \mid x = 0, y + 2z = 0\}$W1 ∩ W2 = {(x, y, z) | x = 0, y + 2z = 0}

No, the answer is incorrect.
Score: 0
Accepted Answers:
$W_2 = \{(x, y, z) \mid 2x + y + 2z = 0\}$W2 = {(x, y, z) | 2x + y + 2z = 0}
$W_1 = \{(x, y, z) \mid x + y + 2z = 0\}$W1 = {(x, y, z) | x + y + 2z = 0}
$W_1 \cap W_2 = \{(x, y, z) \mid x = 0, y + 2z = 0\}$W1 ∩ W2 = {(x, y, z) | x = 0, y + 2z = 0}

Numerical Answer Type (NAT):

Let θθ be the angle between the vectors $u = (2, 1, -3)$u = (2, 1, −3) and $v = (5, 8, 4)$v = (5, 8, 4) and $||.||||.||$ denote the usual length in R^3 R3 with respect to the dot product, then what is $||u||||v||cos(\theta)$||u||||v||cos(θ)?

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 6
1 point
Let $\langle ., . \rangle$⟨., .⟩ denote the standard inner product on R^3 R3, i.e., $\langle (x_1, x_2, x_3), (y_1, y_2, y_3) \rangle = x_1 y_1 + x_2 y_2 + x_3 y_3$⟨(x1, x2, x3), (y1, y2, y3)⟩ = x1y1 + x2y2 + x3y3. Let v_1 = v1 =(1,0,-1), $v_2 = (a, b, c) \in R^3$ v2 = (a, b, c) ∈ R3. Suppose the length of the vector v_2 v2 is 3 and $\langle v_2, (1, -1, 1) \rangle = -3$⟨v2, (1, −1, 1)⟩ = −3, $\langle v_2, (-1, 2, 1) \rangle = 1$⟨v2, (−1, 2, 1)⟩ = 1. If the angle between v_1 v1 and v_2 v2 is 45° 45∘, then find the value of $a + b + c$.a + b + c.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 1
1 point
Let $A_{4\times4}$ A4×4, $B_{4\times4}$ B4×4 and $C_{4\times4}$ C4×4 be matrices such that AA and BB are equivalent matrices and BB and CC are similar matrices. If rank of the matrix AA is 3 then find the determinant of the matrix CC.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 0
1 point